2. Salt-and-Pepper Noise

> *A scanned document contains salt-and-pepper noise affecting readability.*

(a) Scenario and characteristics of the noise

Salt-and-pepper noise (also called impulse noise) appears as randomly scattered black (pepper, value 0) and white (salt, value 255) pixels distributed sparsely across the image, while the surrounding pixels remain unaffected. In a scanned document, this typically arises from dust particles, scanner sensor faults, transmission errors, or damaged/aged paper. Each affected pixel is set to an extreme value independent of its neighbours, unlike Gaussian noise which perturbs every pixel by a small random amount.

(b) Key issues impacting image quality

• Random black/white specks reduce the clarity and readability of printed text.

• The extreme pixel values can be visually confused with actual text strokes or punctuation, degrading OCR accuracy.

• Noise pixels corrupt fine character edges, making character boundaries irregular.

• Simple averaging-based smoothing would blur text further without fully removing the noise, since it mixes the extreme noise values into neighbouring pixels instead of eliminating them.

(c) Applying an appropriate filtering technique

A median filter is applied. For each pixel, a neighbourhood window (e.g. 3x3) is considered, all pixel values in the window are sorted, and the center pixel is replaced with the median value of that neighbourhood. Since salt-and-pepper noise produces extreme outlier values, the median (rather than the mean) is unaffected by these outliers as long as they don't dominate more than half the window.

(d) Justification for this method over alternatives

Unlike a mean (averaging) filter, which blends the extreme noise values into the surrounding pixels and blurs text edges, the median filter is a non-linear, order-statistic-based filter that completely discards outlier extreme values rather than averaging them in. This makes it especially effective against impulsive noise while preserving sharp edges - critical for maintaining the legibility of text strokes in a scanned document. Gaussian smoothing, by contrast, is better suited to additive noise that affects all pixels gradually, not isolated extreme-value impulses.

(e) Summary

Salt-and-pepper noise introduces isolated extreme-value pixels that harm document readability; the key issue is that these outliers corrupt text edges and can be mistaken for genuine content; and the median filter is the most suitable solution because it removes outlier pixels directly while preserving edge sharpness, which averaging-based filters cannot do as effectively.